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MILANO 1863

LiDAR-Aided Cooperative Localization and Environmental Perception for CAVs

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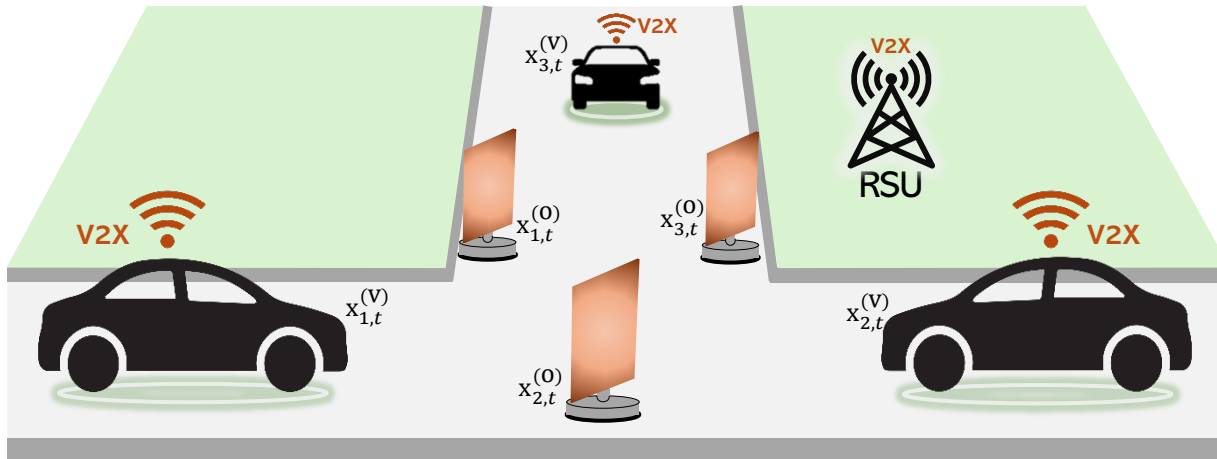


Outline

- Implicit Cooperative Positioning
- Bayesian recursion for NLOS compensation
- Grid-Based Spatial Detection
- Validation in Real World

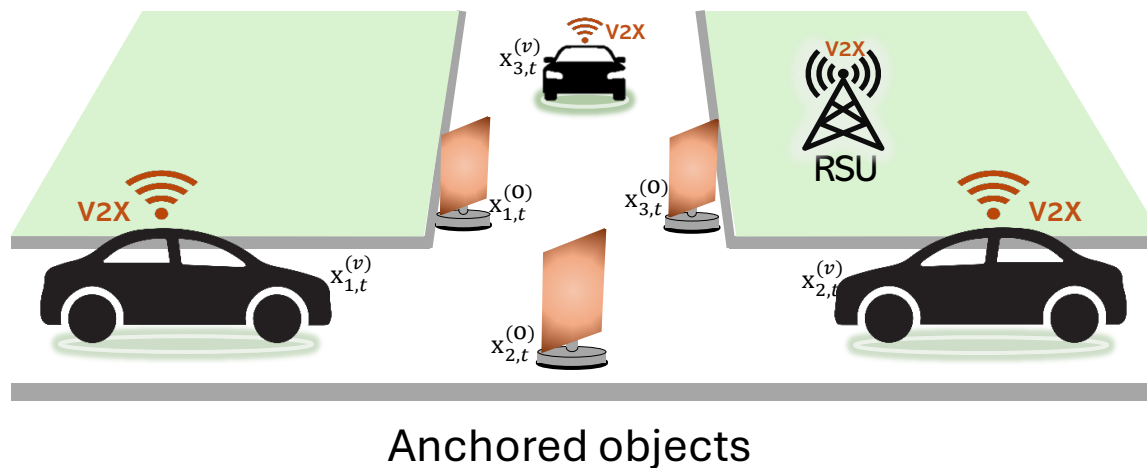
Implicit Cooperative Positioning

- » Relies on exchanging partially processed data
- » Main objective is to localize vehicles
- » Efficient and sufficient





LiDAR-Aided Implicit Cooperative Positioning



State definition

$$\mathbf{x}_i^{(v)} = [u_{i,x}, u_{i,y}, \dot{u}_{i,x}, \dot{u}_{i,y}]^T$$

$$\mathbf{x}_j^{(o)} = [u_{j,x}, u_{j,y}]^T$$

$$\mathbf{x}_{fused} = [\mathbf{x}_1^{(v)}, \dots, \mathbf{x}_{N_v}^{(v)}, \mathbf{x}_1^{(o)}, \dots, \mathbf{x}_{N_o}^{(o)}]^T$$

$$\mathbf{C}_{fused} = \text{blkdiag}(\mathbf{C}_1^{(v)}, \dots, \mathbf{C}_{N_v}^{(v)}, \mathbf{C}_1^{(o)}, \dots, \mathbf{C}_{N_o}^{(o)})$$

Motion dynamics modelling

$$\mathbf{x}_{i,t}^{(v)} = \mathbf{F}\mathbf{x}_{i,t-1}^{(v)} + \mathbf{G}\mathbf{n}_{i,t-1}$$

$$\mathbf{x}_{j,t}^{(o)} = \mathbf{x}_{j,t-1}^{(o)} + \mathbf{w}_{j,t-1}$$

$$\mathbf{F} = [\mathbf{I}_2, \mathbf{T}_s \mathbf{I}_2; \mathbf{0}_{2 \times 2}, \mathbf{I}_2]$$

$$\mathbf{G} = [0.5\mathbf{T}_s \mathbf{I}_2; \mathbf{T}_s \mathbf{I}_2]$$

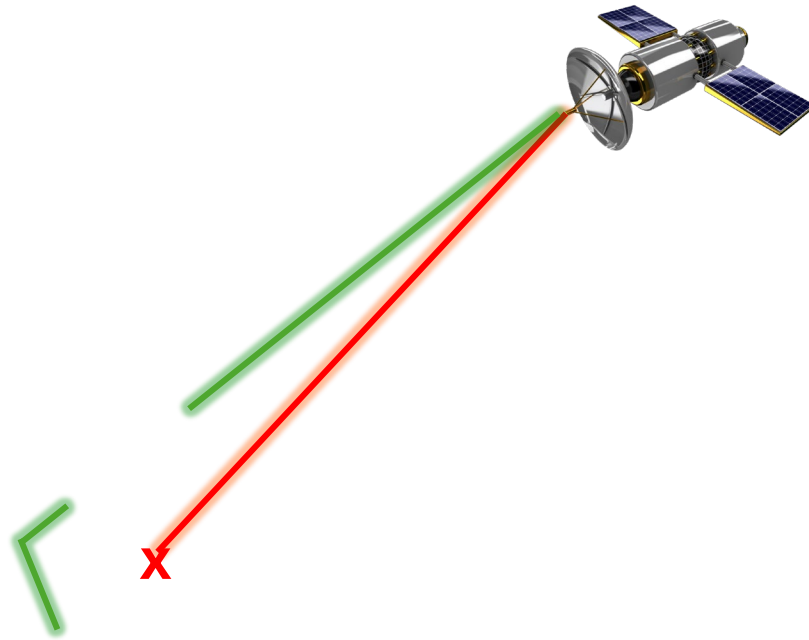
Measurement modelling

$$\boldsymbol{\rho}_{i,t}^{(GNSS)} = \mathbf{x}_{i,t}^{(v)} + \mathbf{q}_{i,t}$$

$$\boldsymbol{\rho}_{i,j,t}^{(OD)} = \boldsymbol{\rho}_{i,t}^{(GNSS)} - \mathbf{x}_{j,t}^{(OD)} + \mathbf{p}_{i,j,t}$$

Linear

Bayesian recursion with NLOS Compensation



KF

$$\mathbf{s}_i^{(v)} = \left[x_i^{(v)}, b_{i,x}, b_{i,y} \right]^T$$

Prediction

$$\mathbf{s}_{i,t}^{(v)} = \mathbf{K} \mathbf{s}_{i,t-1}^{(v)} + \mathbf{P} \bar{\mathbf{n}}_{i,t-1}$$

$$\left\{ \begin{array}{l} \mathbf{K} = [\mathbf{F} \mathbf{O}_{4 \times 2}; \mathbf{0}_{2 \times 4} \mathbf{I}_2] \\ \mathbf{P} = [\mathbf{L} \mathbf{O}_{4 \times 2}; \mathbf{0}_{2 \times 2} \mathbf{I}_2] \end{array} \right.$$

Update

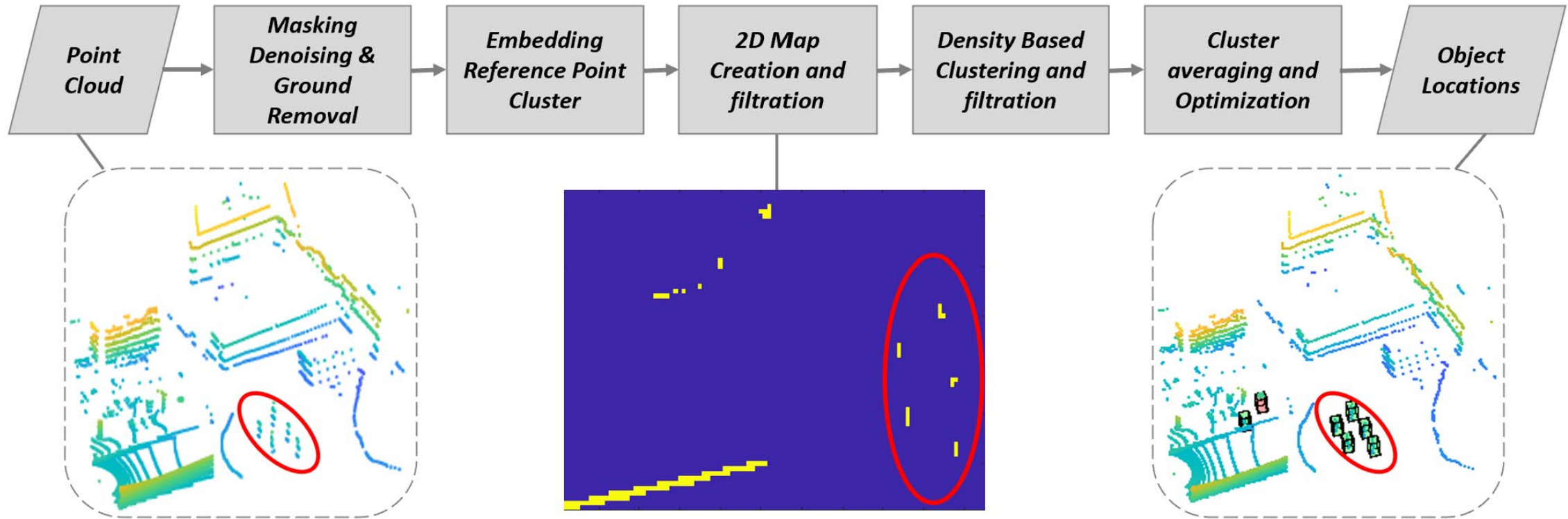
$$\tilde{\rho}_{i,t} = \rho_{i,t}^{(GNSS)} - \hat{\mathbf{b}}_{i,t}$$

MMSE

$$\int x_t p(.) dx_t \rightarrow \hat{\mathbf{s}}_i^{(v)}$$

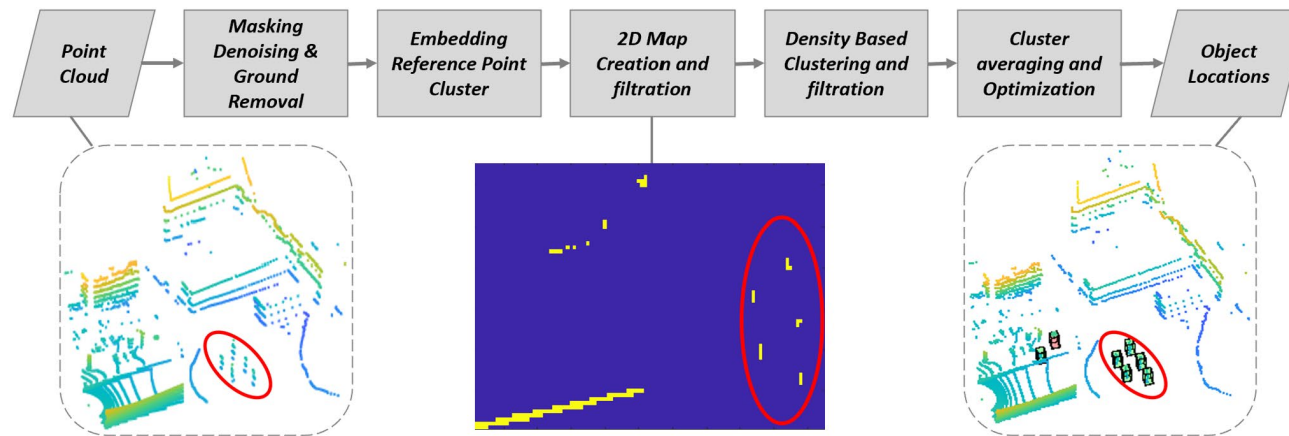
Grid-based Spatial Detection

- » Density-based object detection methods are promising alternatives to deep learning-based object detectors
- » They work well even for lower-resolution LiDARs

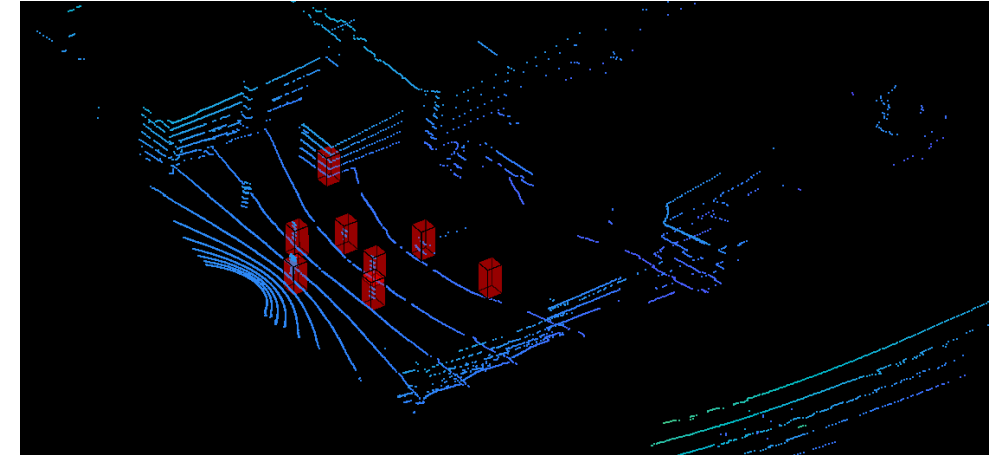


Grid-based Spatial Detection

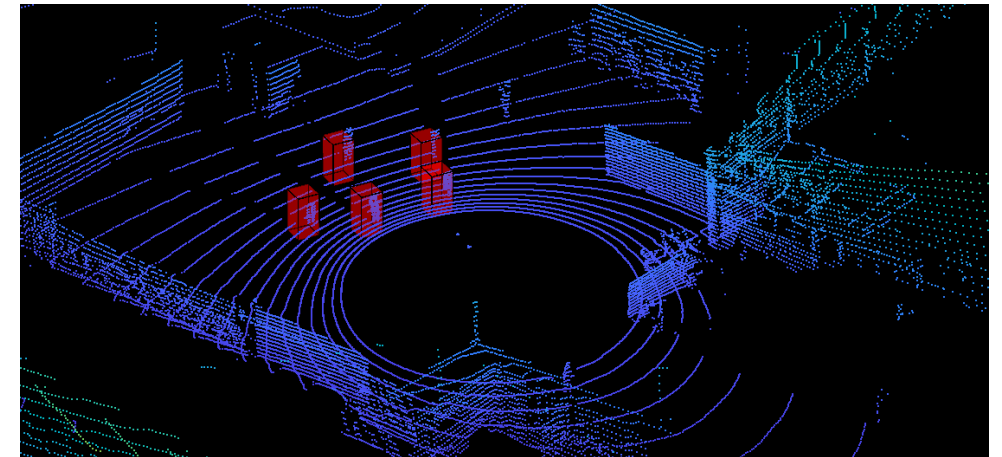
- » Density-based object detection methods are promising alternatives to deep learning-based object detectors
- » They work well even for lower-grade lidars



PREDICTIONS
16 planes lidar



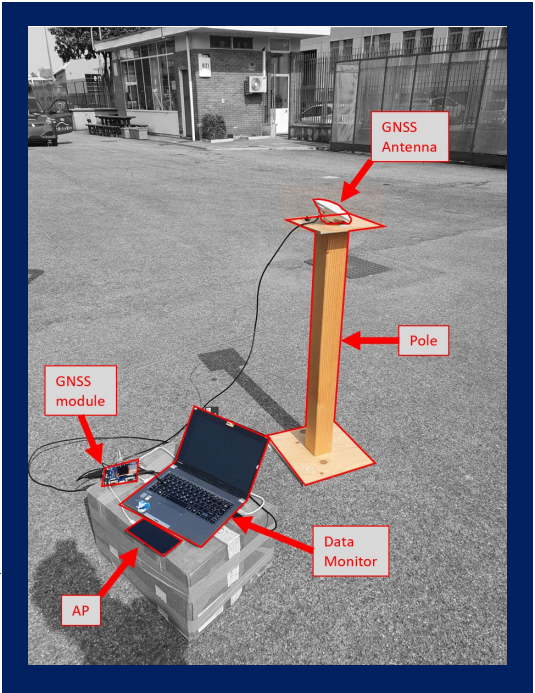
PREDICTIONS
32 planes lidar



Assessment in real world

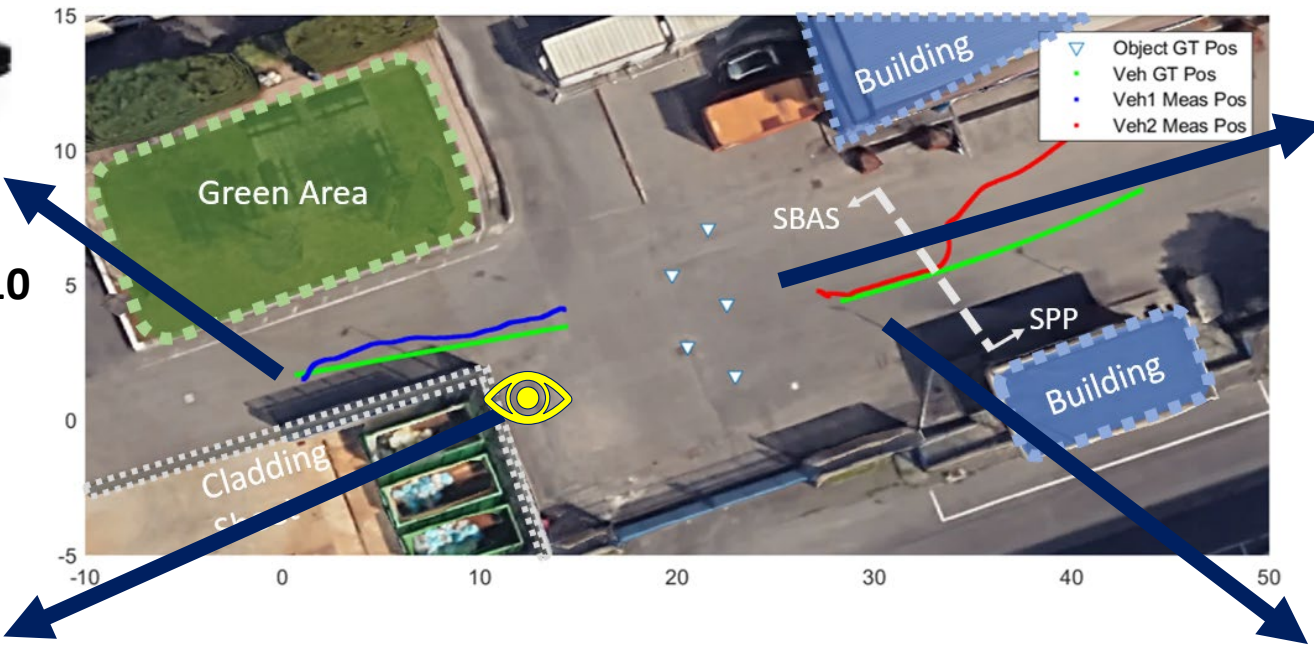


Duration	4 min
# of objects	5
# of vehicles	2



Pole Location Measurement With GNSS-RTK Via IP

TeinVein



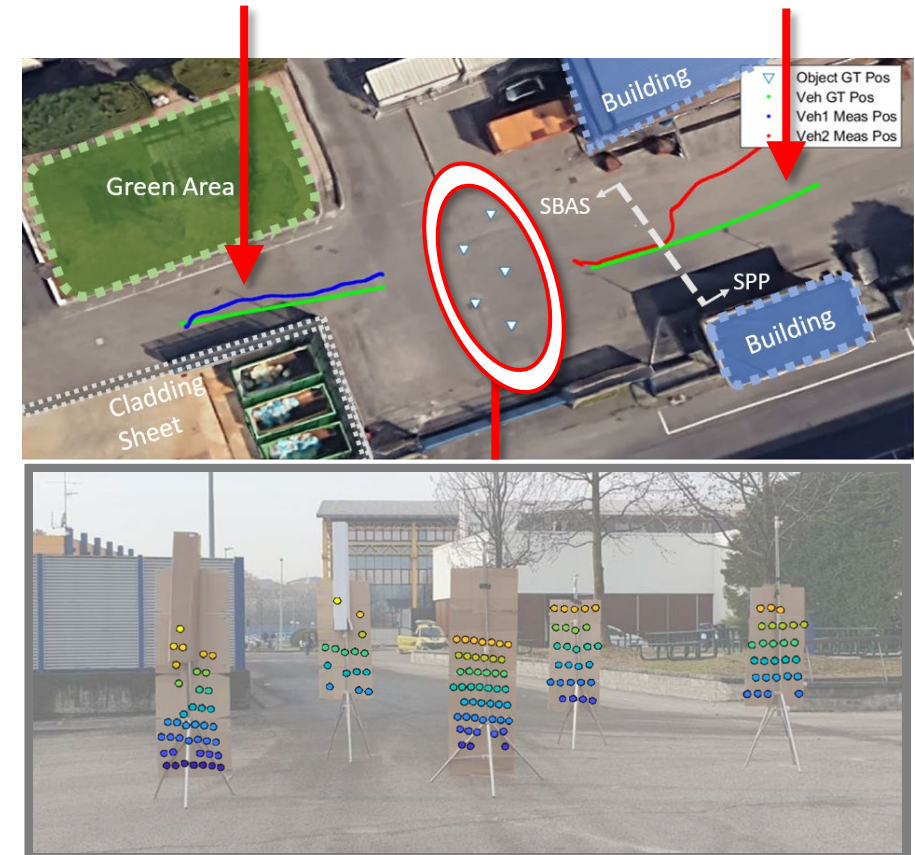
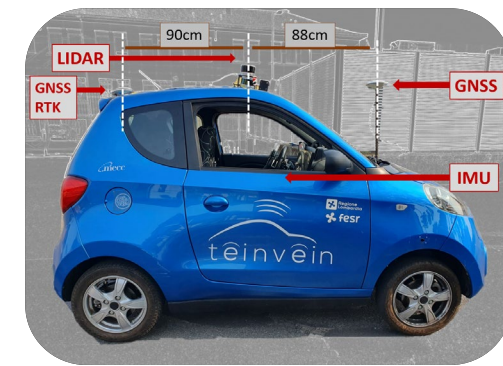
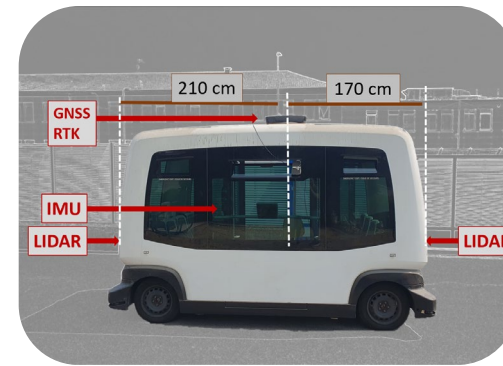
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Realistic experimental campaigns

- Exclusive analysis in urban area featuring GNSS denied region, green area, cladding metal sheet and large square
- Multi-vehicle scenario with different instruments
- **Heterogenous sensing** with different resolutions of LiDAR sensors and accuracies of GPS

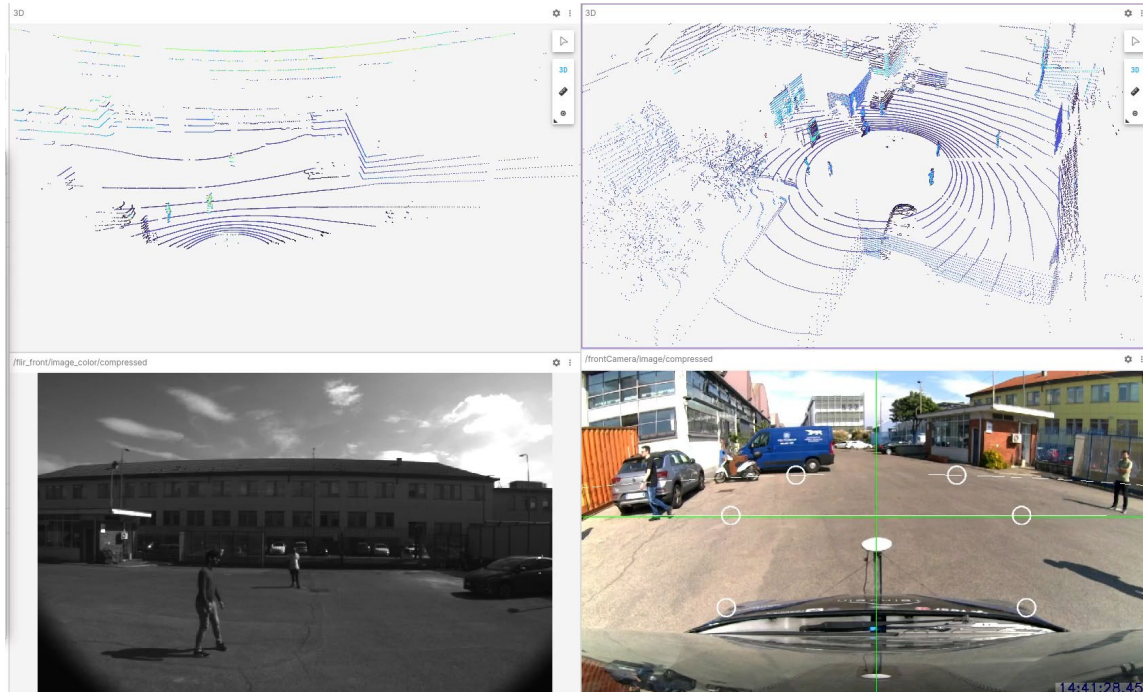
Sensor Type	BUS	CAR
LIDAR	16-Planes	32-Planes
CAMERA	Monochrome	RGB
GNSS RTK	High Accuracy	High Accuracy
GNSS	Low Accuracy	Low Accuracy
IMU	Not exist	Uncalibrated



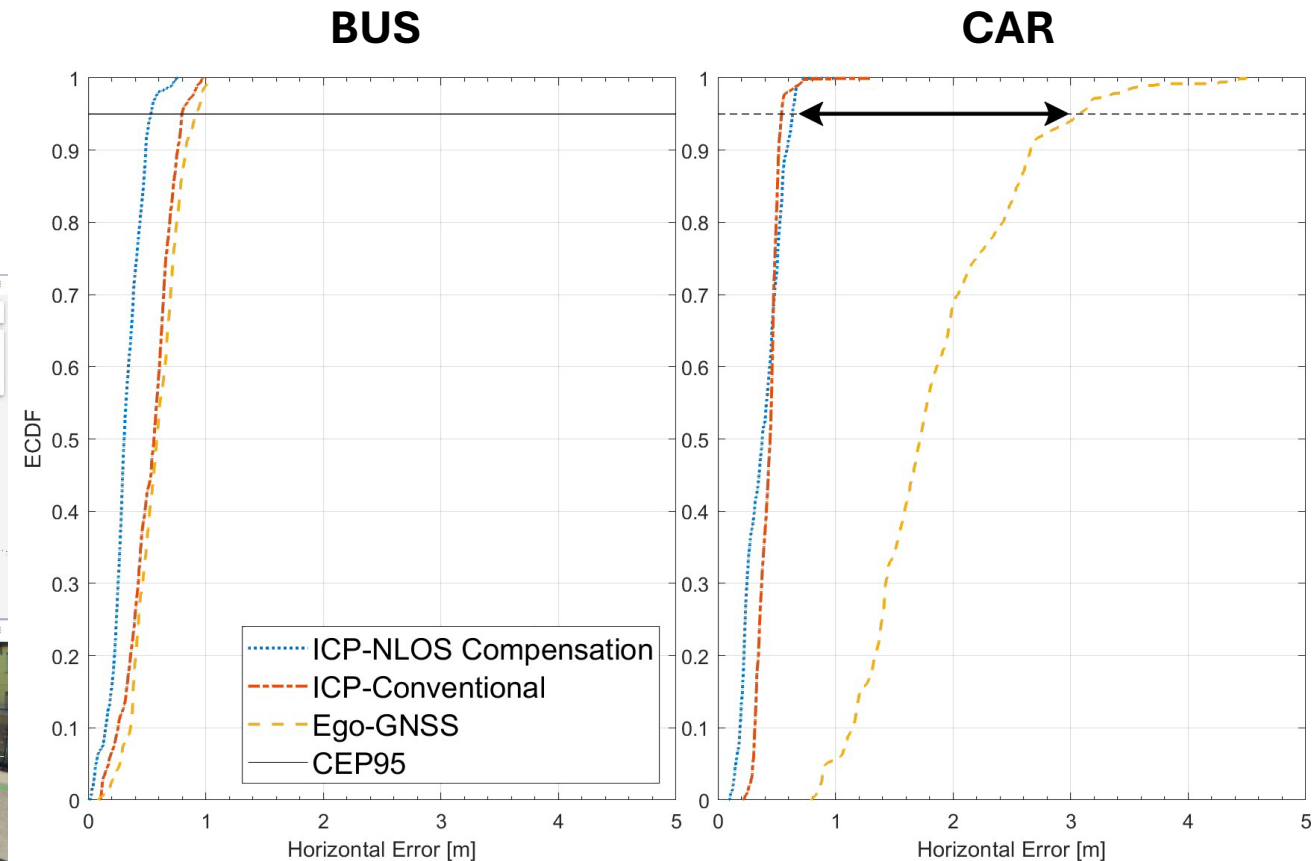
An imagery from static object test

Quantitative assessment of developed algorithms

- Cooperative localization exhibits superior positioning accuracy compared to ego vehicle methods, owing to its ability to harness collaborative information from multiple sensors and entities in dynamic environments



A footage from dynamic object test



GNSS-EKF: CEP95 = 0.83 m

GNSS-EKF: CEP95 = 3.38 m

GNSS-ICP: CEP95 = 0.51 m

GNSS-ICP: CEP95 = 0.57 m

Conclusion

Vehicles with high positioning and object detection accuracy refine localization of vehicles in close proximity. This can be extended by including RSU in this ecosystem.

On-board sensors widely used for automated driving system can be further harnessed for cooperative sensing and positioning to achieve more accurate global positioning.

Future Work

This method relies on several assumptions such as perfect object association and precise vehicular heading calculation. Future work is to increase feasibility by replacing these assumptions with solutions such as multi-Bernoulli filtering and INS calibration.





Drive Safe, Stay Safe

